

Binomial Distribution

The **binomial distribution** is the theoretical probability distribution appropriate when modeling the expected outcome, X , of N trials (or event sequences) that have the following characteristics:

- The outcome on every trial is binary
 - also called a **Bernoulli trial**
- The probability of the target outcome (usually called a “success”) is the same for all N trials
- The outcomes of the trials are *independent*
 - The probability of a success in any one trial must be the same from trial to trial
- The number of trials is fixed (you know how many times you're flipping a coin)

If these assumptions hold then X is a binomial random variable representing the *expected number of successes* over N trials, with expected success on each trial of θ .

A common and compact way of stating the same thing is:

$$X \sim B(N, \theta)$$

Note: Last lecture we used p to denote the probability of success. This time we'll use θ . θ is more correct for the population parameter, but you'll see p used a lot, too.

The probability distribution for X is defined by the following **probability mass function**:

$$P(X|\theta, N) = \frac{N!}{X!(N - X)!} \theta^X (1 - \theta)^{N - X}$$

The probability mass function tells us what to expect for any particular value of X in the sample space.

The probability distribution for X is defined by the following **probability mass function**:

$$P(X = k | n, p) = \binom{n}{k} p^k (q)^{n-k}$$

This is the same exact thing as previous slide!

All theoretical distributions have a mass function (if discrete) or a density function (if continuous). These are the defining equations that tells us the generating process for the behavior of X .

$$P(X|\theta, N) = \frac{N!}{X!(N-X)!} \theta^X (1 - \theta)^{N-X}$$

$P(X|\theta, N)$ is a conditional probability: the probability of X **given** θ and N .

- X is the number of successful trials (in our other formula this is k -- as in r correct "choices")
- N is the number of trials; must be independent
- θ is the probability of success on any given trial (in our other formula, this is p)

θ and N are parameters of the binomial distribution.

The probability mass function tells us what to expect for any particular value of X in the sample space.

$$P(X|\theta, N) = \frac{N!}{X!(N-X)!} \theta^X (1 - \theta)^{N-X}$$

$\theta^X (1 - \theta)^{N-X}$ is the probability of any particular instance of X .

- This is just a general form of the basic probability rule:

$$A \text{ and } B = P(A \cap B) = P(A)P(B)$$

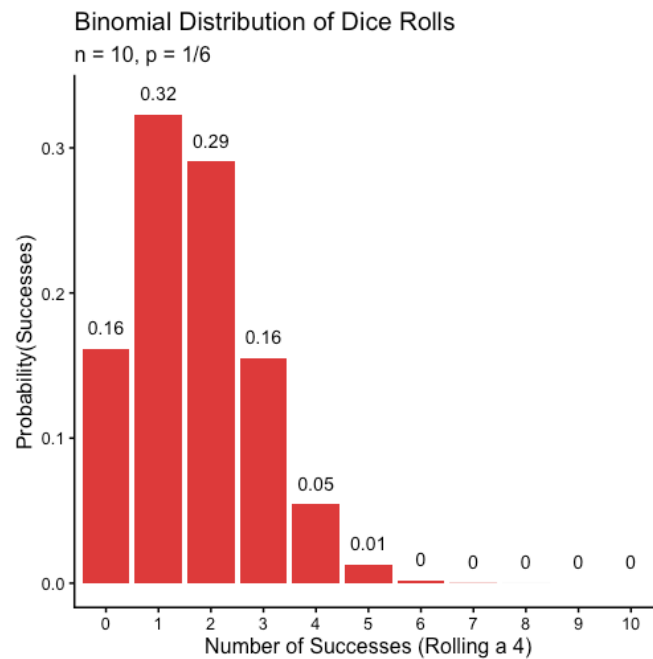
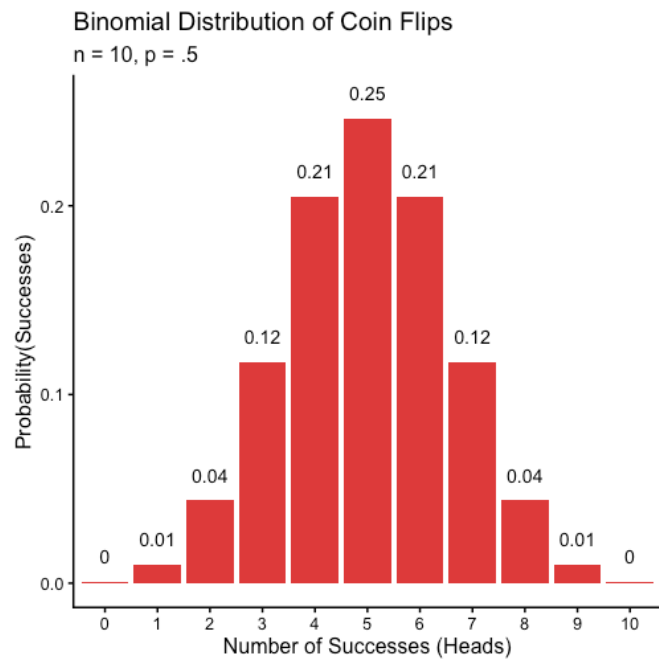
- AKA the intersection AKA $p(\text{success}) \times p(\text{failure})$
- Note that this form of the rule assumes *independent events*.

Binomial Distribution

$$p(X = k); n, p = \binom{n}{k} p^k (1 - p)^{n-k}$$

- Parameters of this distribution are n and p
- If we change the parameters, we change the distribution
- **Family** of distributions

Family of Distributions



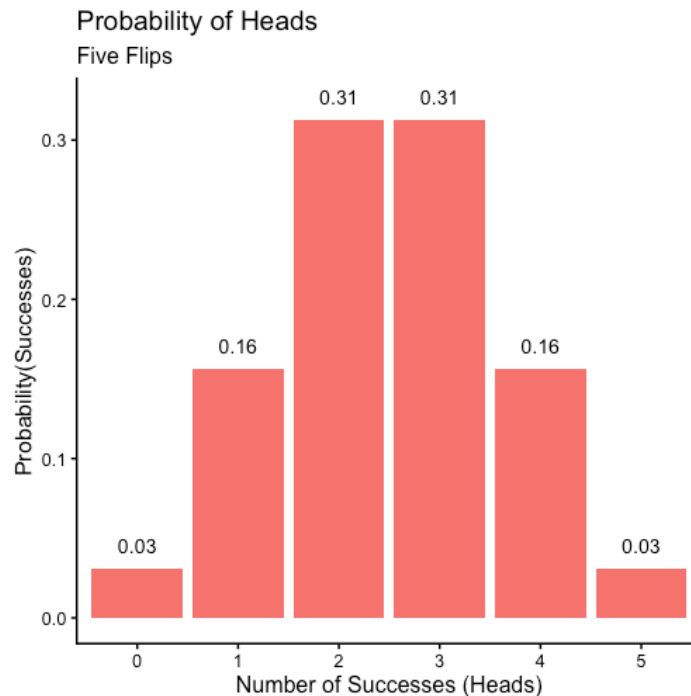
Parameters

Every probability distribution has an **expected value**.

- Expected Value is essentially the average value if you repeated the experiment...a lot
- Most likely result of the probability function
- The thing we would expect to happen if we have no other information than the parameters of the distribution
- The *long-run* average over an infinite amount of trials or samples

Averages vs. Expected Value

- What is the average number of successes out of 5 flips?
- In 5 coin flips, what is the average number of times you'd get "heads"?



Parameters

Every probability distribution has an **expected value**.

For the binomial distribution:

$$E(X) = N\theta$$

For 3 Heads out of 5 flips, $E(X) = 5 \times .5 = 2.5$

Each probability distribution also has a variance. For the binomial:

$$Var(X) = N\theta(1 - \theta)$$

Importantly, this means our mean and variance are related in the binomial distribution, because they both depend on θ . How are they related?

Let's set our mean = variance

$$N\theta = N\theta(1 - \theta)$$

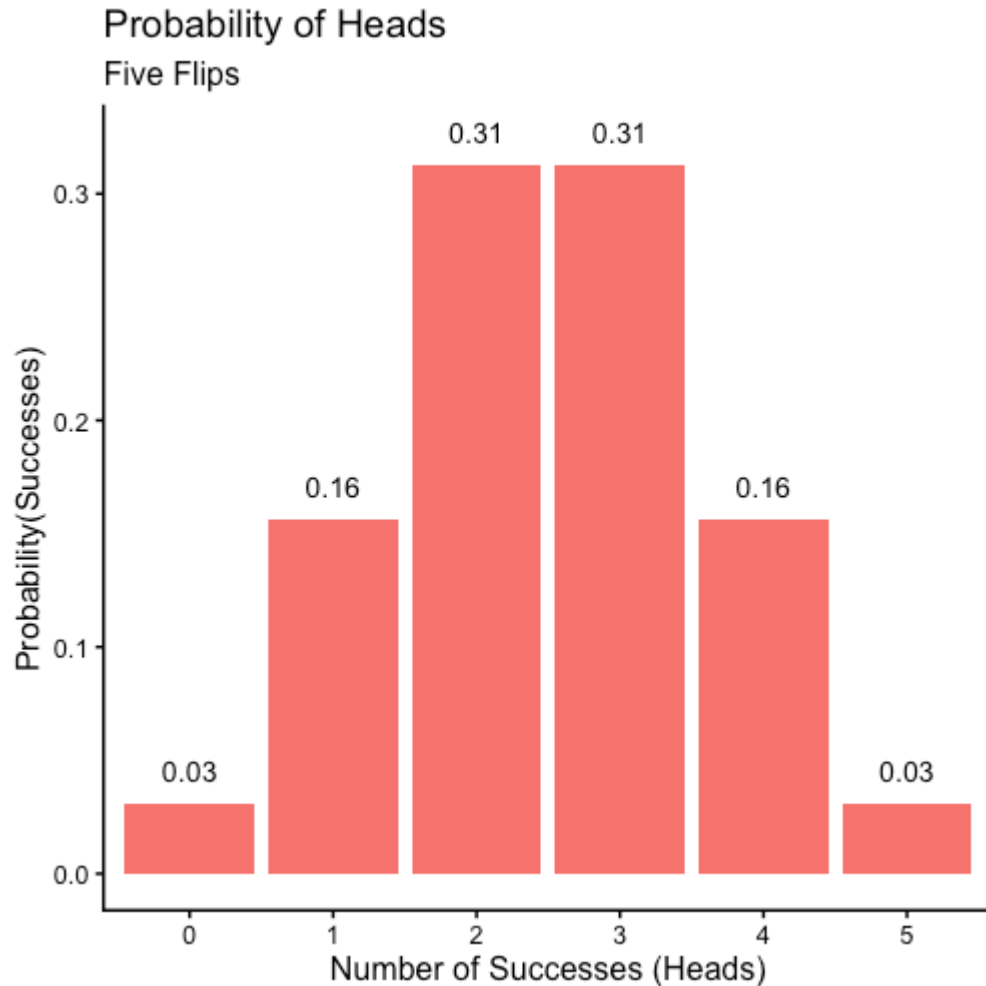
$$Np = Np(1 - p)$$

$$Np = Np(q)$$

$$\text{mean} = \text{mean} \times q$$

- If $q = 1$, then $E(X) = Var(X)$
- Else, $E(X) > Var(X)$

The mean, 2.5, does not exist in the sample space, and rounding up to 3 and claiming that to be the most typical outcome isn't right either.



- If you have a discrete distribution with a small N , these estimates may not have a sensible meaning

The **probability mass (density) function** allows us to answer other questions about the sample space that might be more important, or at least realistic.

- mass = discrete
- density = continuous

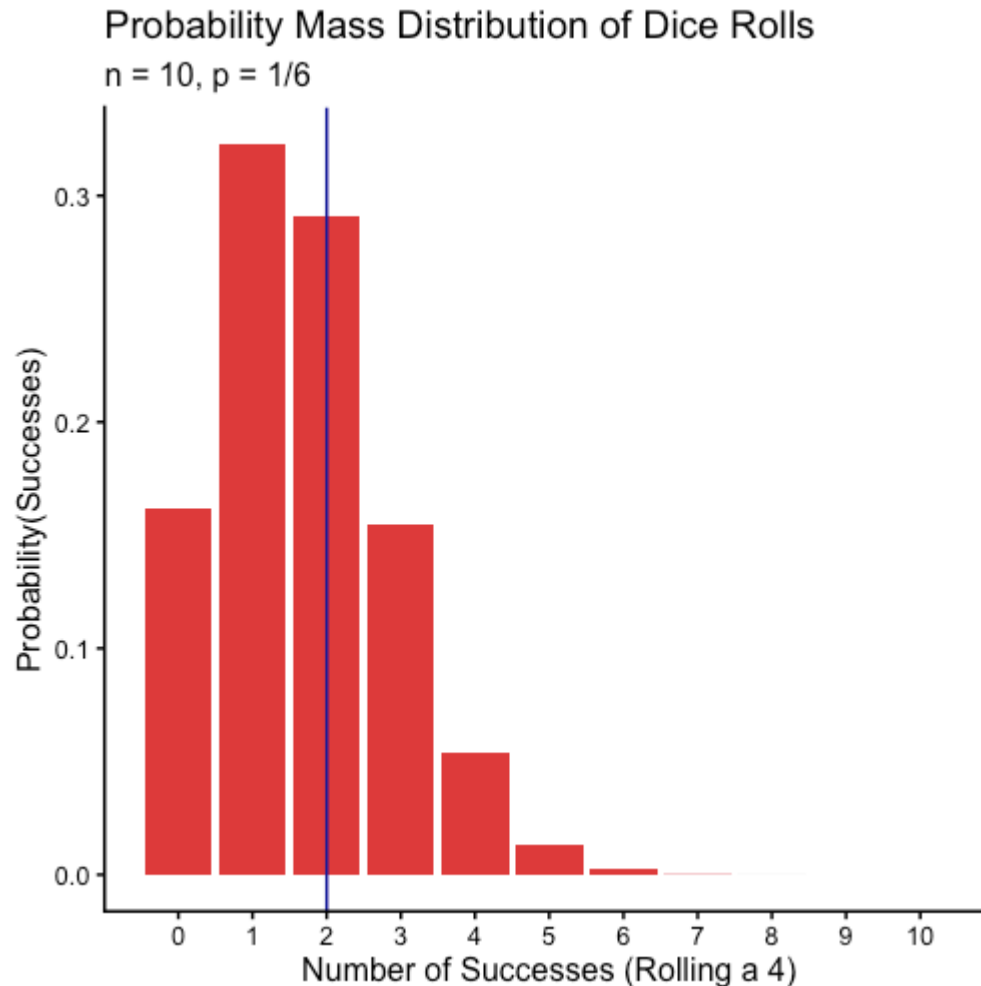
I might want to know the value in the sample space at or below which a certain proportion of outcomes fall.

"At or below what outcome in the sample space do 75% of the outcomes fall?"

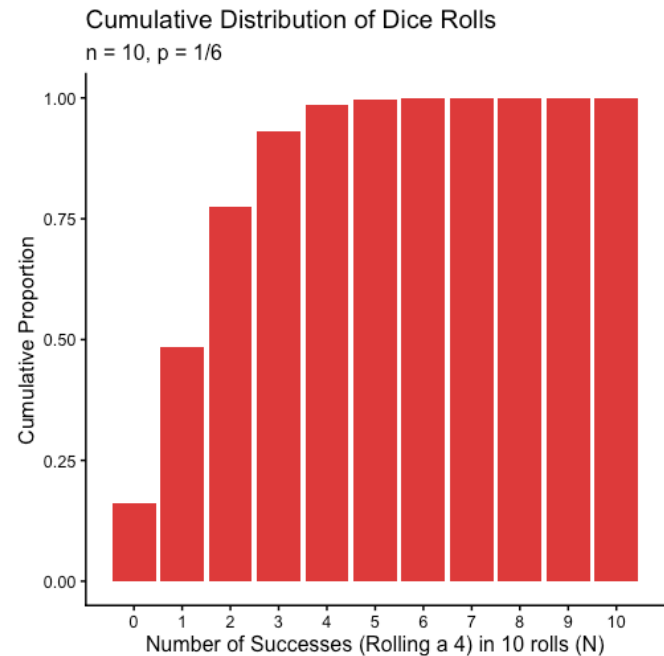
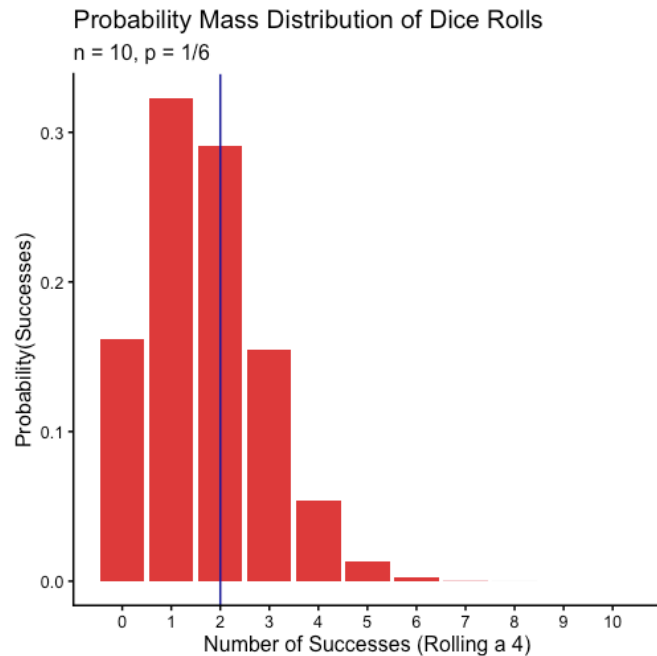
This is a **percentile or quantile** question.

I might want to know the proportion of outcomes in the sample space that fall at or below a particular outcome. This is a **cumulative proportion** question.

At or below what outcome in the sample space do 75% of the outcomes fall? What is the outcome? Providing the 75%, trying to find the 2...



What proportion of outcomes in the sample space that fall at or below a given outcome? What percentage of outcomes fall at or below 2? Providing the 2, looking for the 75%

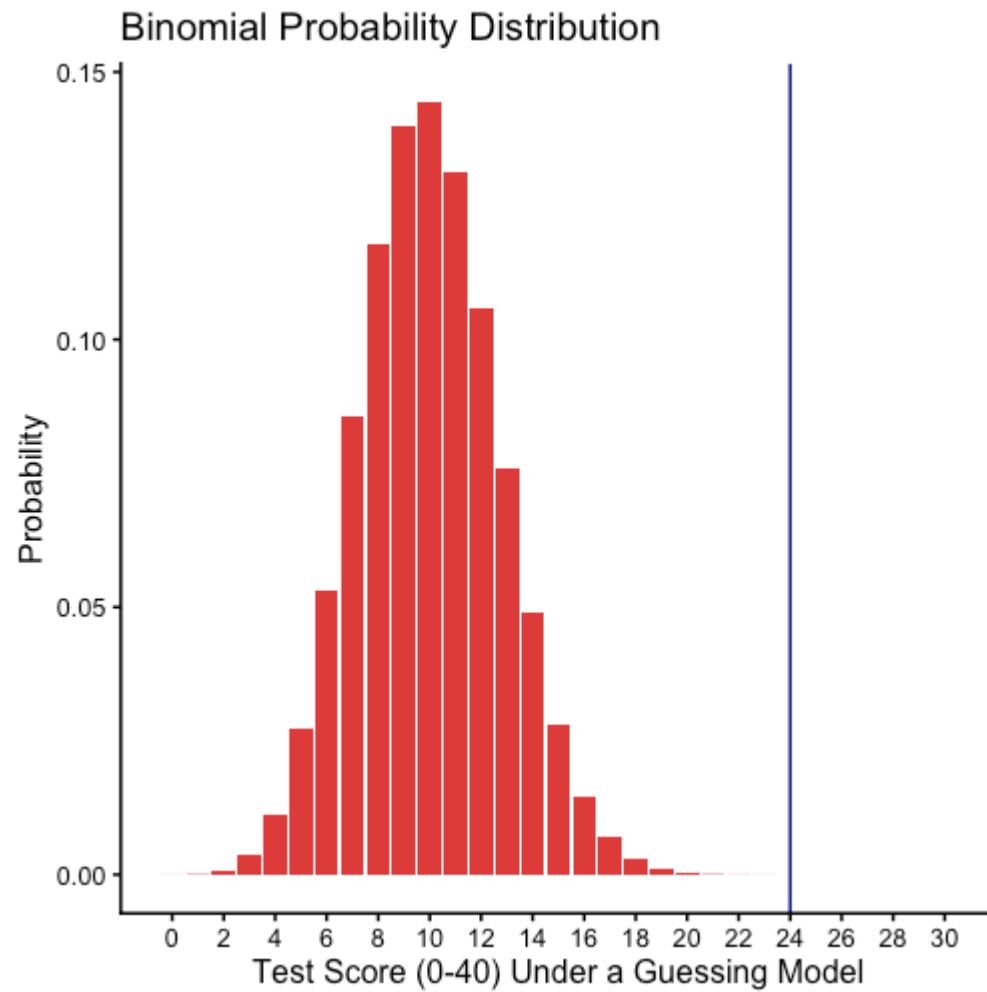


The binomial is of interest beyond describing the behavior of dice and coins.

Many practical outcomes might be best described by a binomial distribution.

For example, suppose I give a 40-item multiple choice test, with each question having 4 options.

- I am worried that students might do well by chance alone. I would not want to pass students in the class if they were just showing up for the exams and guessing for each question.
- What are the parameters in the binomial distribution that will help me address this question?
 - $N = 40$
 - $\theta = .25$



By hand

How likely is it that a guesser would score above the threshold (60%) necessary to pass the class by the most minimal standards?

$$\begin{aligned} &P(24|.25, 40) + \\ &P(25|.25, 40) + \\ &P(26|.25, 40) + \\ &\quad \dots \\ &P(40|.25, 40) \end{aligned}$$

In R

How likely is it that a guesser would score above the threshold (60%) necessary to pass the class by the most minimal standards?

In R, we can calculate the cumulative probability (X or lower), using the `pbinom` function.

The probability of getting 23 questions or fewer correct:

```
pbinom(q = 23, size = 40, prob = .6)
```

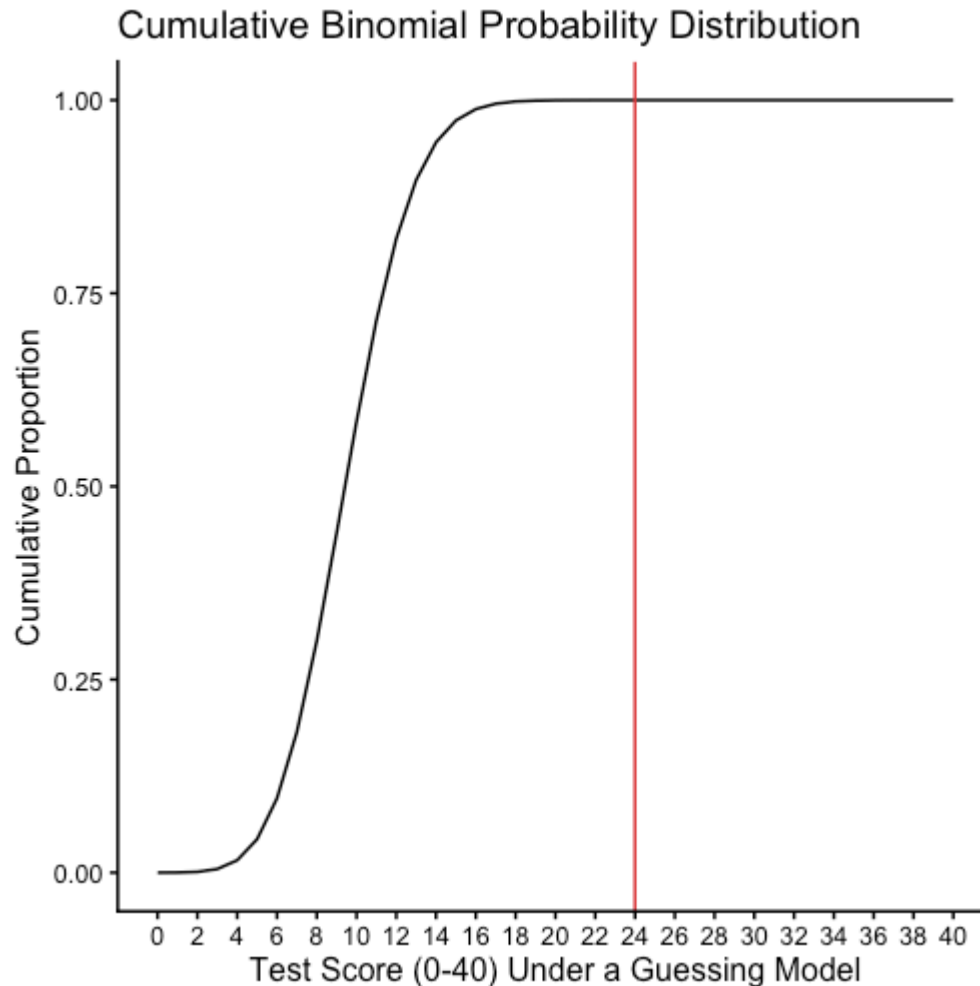
```
## [1] 0.9999972
```

The probability of getting 24 or more questions correct:

```
#Note the use of the Law of Total Probability  
1-pbinom(q = 23, size = 40, prob = .6)
```

```
## [1] 2.825967e-06
```

Cumulatively, what proportion of guessers will fall below each score?



There's always a "but"

But, what assumptions are we making and what consequences will they have?

- The outcome on every trial is binary (also called a Bernoulli trial)
- The probability of the target outcome (usually called a "success") is the same for all N trials
- The trials are **independent**
$$P(A \cap B) = P(A|B)P(B) = P(A)P(B)$$
- The number of trials is fixed

In probability and statistics, if the assumptions are wrong then inferences based on those assumptions could be wrong too, perhaps seriously so.

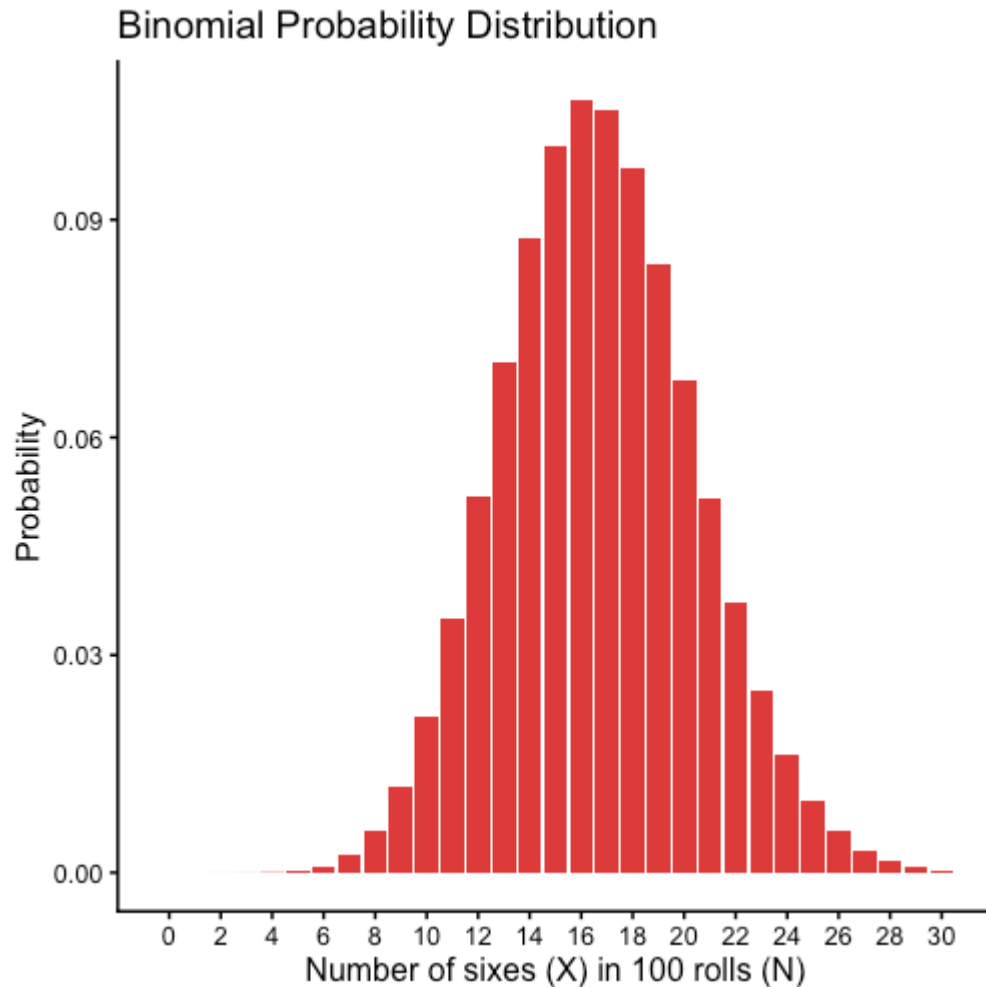
All models are wrong, but some models are useful.
(G.E.P. Box)

We might have viable alternative models:

- **Geometric distribution:** Used if we are interested in the number of trials required for one "success" to occur
 - "how many times do I start my computer before it fails to start at all?"
- **Negative binomial distribution:** Used if we are interested in the number of successes in a series of repeated trials until a specified number of failures are seen
 - "A child won't return from trick or treating until they get 5 full-size candy bars. What is the probability that they will have to visit 34 homes to get this?"

As N increases, the binomial becomes more normal in appearance.

Because of the difficulties in calculating large factorials, there is a large-sample normal approximation to the binomial. The normal distribution is useful for a lot of other reasons too.



Next time...

the normal distribution